

Kouds HALITIM

PhD student in control of HPC systems

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SKILLS

Research	Feedback control in distributed infrastructures, Thermal-aware resource management, Congestion mitigation in storage systems, Hybrid control and scheduling frameworks, Performance Modelling and Predictions, Sustainable computing, Digital sobriety, Energy efficiency, Instrumentation and Monitoring Tools, Auto-Tuning and Portability
Programming	C, C++, Python, MATLAB, R
Databases	MySQL
Tools & DevOps	Git, Docker, Make, Nix, VS Code, oar
Operating Systems	Windows 11, Linux Ubuntu

WORK EXPERIENCE

Present Nov. 2023	PhD Candidate in Control of HPC systems, INRIA Ctrl-A Team, Inria, LIG, Univ. Grenoble Alpes, Grenoble, France
Sept. 2023 March 2023	Research Intern, INRIA Spirals Team, Inria, Lille, France
Jan. 2023 Dec. 2022	Industrial Automation Intern, Sonatrach Sonatrach, Skikda, Algeria
March 2022 April 2022	Embedded Systems Intern, Brandt Brandt, Sétif, Algeria
April 2021 March 2021	Instrumentation Intern, IRIS IRIS, Sétif, Algeria

EDUCATION

November 2026 (Expected)	PhD in Computer Science, INRIA Ctrl-A Team, Inria, LIG, Univ. Grenoble Alpes, Grenoble, France. "Efficient task hybridization in heterogeneous computing : a practical combination of control and scheduling theories." Supervised by : Eric Rutten, Raphael Bleuse, and Bogdan Robu. ➤ Developed models for large-scale distributed systems (HPC) to account for system dynamics, resource constraints, and thermal uncertainties. ➤ Synthesized and implemented controllers to regulate trade-offs between file system congestion, resource harvesting and power consumption. ➤ Designed a Stochastic Model Predictive Control framework to optimize task injection under high uncertainty and variable workload conditions. ➤ Formulated I/O interference as a control problem and implemented regulation mechanisms for multi-client environments to ensure performance isolation. ➤ Built large-scale experimental testbeds and automated data collection to validate theoretical models against state-of-the-art heuristics. ➤ Conducted robustness analyses and comparative evaluations to assess the stability, performance and scalability of implemented controllers. ➤ Designed a hierarchical regulation framework for processor power capping that ensures energy budget compliance across variable system dimensions and hardware configurations.
September 2023	Master's and Engineering Degree, ENP National Polytechnic School, Algiers, Algeria.

TEACHING

June 2026 September 2024	Discrete-Time Control Systems, Lab Sessions - 18 hours ➤ Developed comprehensive lab curricula and source code for a digital motion control lab, integrating micro-controllers with real-time data processing. ➤ Mentored diverse student groups (15-20) by applying theoretical control concepts to real-world experimental scenarios. ➤ Collaborated proactively with the teaching team to improve educational materials.
June 2026 September 2024	Continuous-Time Control Systems, Lab Sessions - 59 hours ➤ Contributed to the drafting and updating of laboratory workbooks. ➤ Created digital supporting resources on MATLAB/Simulink to facilitate the adoption of calculation and simulation tools. ➤ Commissioned, tested, and configured specific testbeds (PID Controllers and Rotating Machines) and experimental rigs (DC motors, sensors). ➤ Led the full control design process : modeling, experimental identification, and PID synthesis to meet performance criteria.
April 2026	Software Engineering Project, Project Tutor - 5.5 hours ➤ Supervised student teams in the full-cycle development of strategy game software. ➤ Mentored groups on Agile methodologies : task planning, workload distribution, and conflict resolution. ➤ Provided technical oversight on the implementation of Minimax algorithms and heuristic optimization. ➤ Guided the design and validation of Graphical User Interfaces (GUI) using user-centered testing. ➤ Evaluated technical deliverables, including code quality, architectural robustness, and documentation. ➤ Acted as a member of the final defense jury to assess project outcomes and technical presentations.

SUPERVISION

2026	Doaa Elnoamany : “ <i>Control of uncertain MIMO systems with probabilistic constraints...</i> ” Master of Science (M.Sc.) MiSCIT, Université Grenoble Alpes, Grenoble, France. Co-encadrement : Bogdan Robu and Sophie Cerf
2025	Mohamed Abdeljalil Maziz : “ <i>LPV control for dynamic power capping in High-Performance Computing (HPC) under mixed workloads</i> ” Master of Science (M.Sc.) MiSCIT, Université Grenoble Alpes, Grenoble, France. Co-encadrement : Bogdan Robu and Sophie Cerf
2025	Mahmoud Abdo : “ <i>Model Predictive Control (MPC) approach for resource scavenging in High-Performance Computing (HPC)</i> ” Master of Science (M.Sc.) MiSCIT, Université Grenoble Alpes, Grenoble, France. Co-encadrement : Bogdan Robu and Sophie Cerf
2024	Jawad Hasnain : “ <i>Extremum Seeking Control (ESC) for first-order systems with various objective functions</i> ” Master’s (M1) in Electrical Engineering and Control Systems, Université Grenoble Alpes, Grenoble, France. Co-encadrement : Bogdan Robu

PUBLICATIONS

International Conferences

- [C1] **K. Halitim**, T. Collignon, S. Cerf, B. Robu, E. Rutten, R. Bleuse, “*Event-Triggered Adaptive PI Control for Mitigating Congestion in Shared Storage Systems*”, in ECC 2026 - European Control Conference, Reykjavik, Iceland, 2026. (Accepted).
- [C2] **K. Halitim**, S. Cerf, B. Robu, and É. Rutten, “*Robust Cascade Control for Variable Dimension Systems - Application to processors’ power capping*”, in ECC 2025 - European Control Conference, Thessaloniki, Greece, 2025. Available : <https://hal.science/hal-05117585v1>

- [C3] **K. Halitim**, B. Robu, S. Cerf, R. Bleuse, E. Rutten, "Stochastic Model Predictive Control with Direct Feedforward Compensation", in ICSTCC 2025 - 29th International Conference on System Theory, Control and Computing, Cluj-Napoca, Romania, 2025. Available : <https://hal.science/hal-05210549v1>
- [C4] T. Collignon, **K. Halitim**, R. Bleuse, S. Cerf, B. Robu, É. Rutten, L. Seinturier, and A. van Kempen, "Mitigating Shared Storage Congestion Using Control Theory", in UCC 2025 - IEEE/ACM International Conference on Utility and Cloud Computing, Nantes, France, Dec. 2025. Available : <https://cnrs.hal.science/LIG/hal-05368563v1>

National Conferences

- [N1] T. Collignon, **K. Halitim**, R. Bleuse, S. Cerf, B. Robu, É. Rutten, L. Seinturier, and A. van Kempen, "Coarse-grain Congestion Regulation Using Control Theory", in COMPAS 2025 - Conférence en Parallélisme, Architecture et Système, Bordeaux, France, 2025. Available : <https://hal.science/hal-05075480v1>

Theses

- [T1] **Kouids Halitim**. "Enhancing Efficiency through Control Theory in Compute-Intensive Applications," Master's Thesis, 2023. Available : <https://inria.hal.science/hal-04357812>

Posters

- [P1] T. Collignon, **K. Halitim**, R. Bleuse, S. Cerf, B. Robu, É. Rutten, L. Seinturier, and A. van Kempen, "Temperature control of computing sites through the injection of useful and non-invasive tasks", Greendays 2026, March 2026, Sophia, France. Available : <https://inria.hal.science/hal-05551250v1>
- [P2] **Kouids Halitim**. "Task hybridization in heterogeneous computing : a combination of control and scheduling theories", "PAD Days" GIPSA, December 2025, GRENOBLE (38000), France.
- [P3] **Kouids Halitim**. "Task hybridization in heterogeneous computing : a combination of control and scheduling theories", "PhD Madness" LIG, May 2025, GRENOBLE (38000), France. Available : <https://inria.hal.science/hal-05108245v1>

SCIENTIFIC RESPONSIBILITIES

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| 2025 | Reviewer for the European Control Conference (ECC 2025) and the International Conference on Utility and Cloud Computing (UCC 2025) |
| 2025 | Session Chair for the "Control Applications" session at the European Control Conference (ECC 2025) |
| 2026 | Session Chair for the "Heterogeneous and reconfigurable architectures for the future of computing: Project Talks" session at the 18 th JLESC workshop |

LANGUAGES

French : Fluent

English : Fluent

Arabic : Native

German : Beginner